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Optoelectronic resistive random access memory for neuromorphic vision sensors

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Supplementary Information

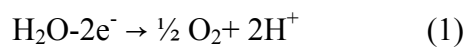
Optoelectronic resistive random access memory for neuromorphic vision sensors

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Supplementary Note 1

The X-Ray diffraction (XRD) pattern (Supplementary Fig. 1a) demonstrates the amorphous structure of the as-deposited MoO_x, which possesses good photochromic properties in terms of fast response and long retention time¹. The optical band gap of as-prepared MoO_x is extracted as 3.37eV from the absorption spectra (Supplementary Fig. 1b).

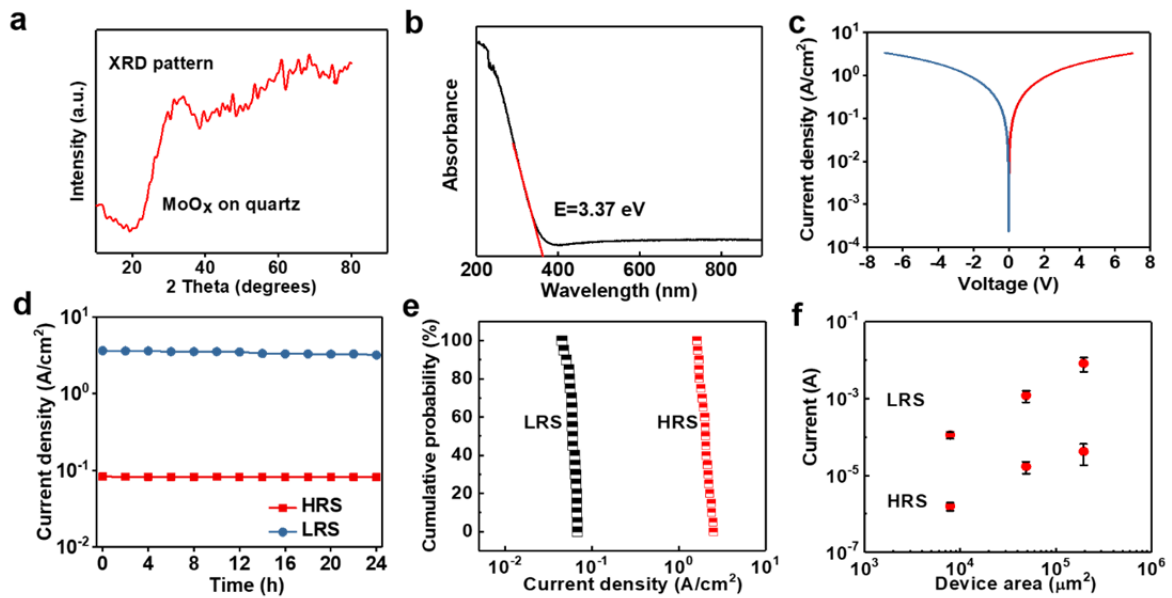
We first perform voltage sweeps from 0 ~ 8 V and 0 ~ -8V without UV illumination, as shown in Supplementary Fig. 1c. No electrical resistive switching behaviour is observed. Although applied voltage can also induce the water dissociation without UV light, no electrical switching is observed. This can be understood as a result of different reactions. The electrically induced water dissociation reactions at cathode and anode can be expressed as:



Under external electric field, H⁺ and OH⁻ are drifted towards each other and then combined. The valence state change of Mo only occurs near the metal/MoO_x interface. The small amount of valence state change of Mo at the interface is insufficient for forming percolation network through MoO_x thin film, resulting in negligible resistance change. In contrast, the

products from UV-induced water dissociation induce H^+ in the MoO_x film, which can produce uniform distribution of H^+ and cause the change of Mo valence state.

Supplementary Fig. 1d shows the retention after the pulse switching. Both HRS and LRS are remained nearly unchanged after 24 hours, exhibiting typical non-volatile characteristics. The cumulative probability plot (Supplementary Fig. 1e) shows small device-to-device variation. The MoO_x ORRAM also exhibits area-dependent switching characteristics (Supplementary Fig. 1f).



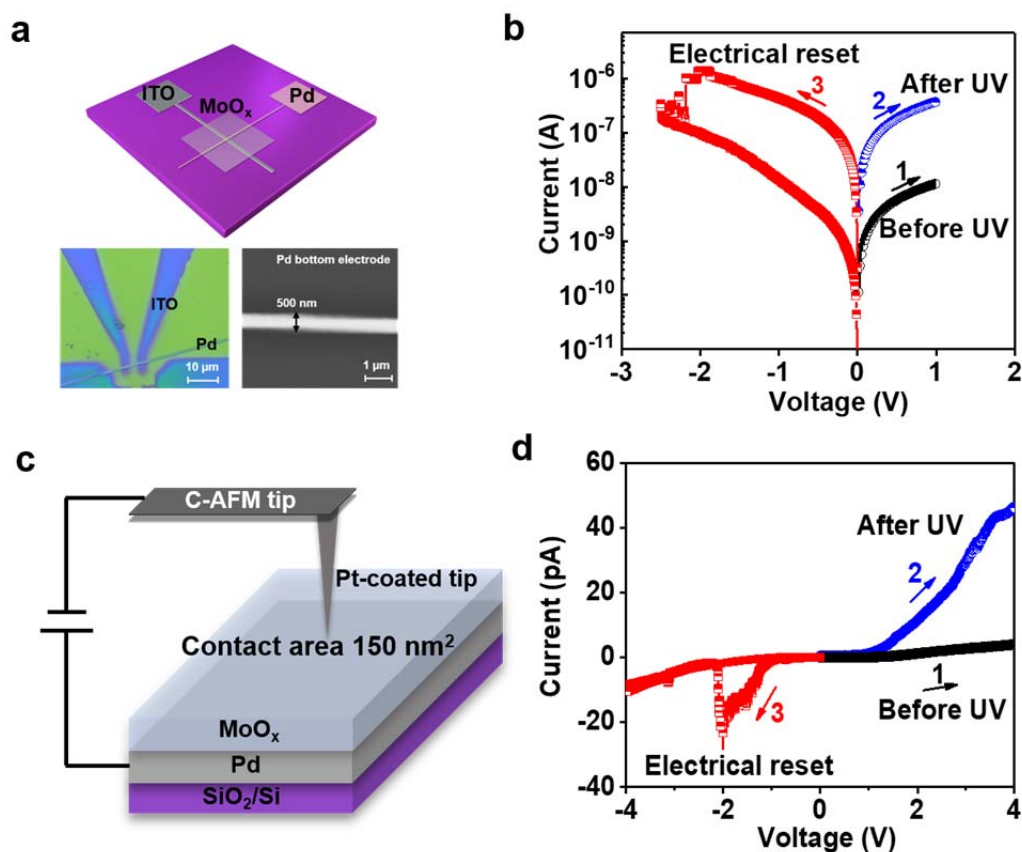
Supplementary Fig. 1. Non-volatile switching characteristics. (a) XRD pattern of MoO_x layer on quartz substrate, suggesting the amorphous structure of the as-deposited MoO_x . (b) Absorption spectrum of as-prepared MoO_x layer on quartz substrate. The bandgap is extracted as 3.37 eV. (c) Current-voltage curve of as-fabricated MoO_x device without UV illumination by sweeping voltage. (d) Retention test of pulse switching. The SET process is operated by a UV light (365 nm) pulse with an intensity of 150 mW/cm² and a pulse width of 600 ms. Electrical pulse (-4.5 V, 100 ms) is used for the RESET process. (e) Cumulative plot of resistances at LRS and HRS of 20 devices at different batches. (f) Area-dependent switching characteristics. The current levels at both HRS and LRS increase with the increase of device area, excluding the possibility of filamentary switching mechanism.

Supplementary Note 2

In order to confirm the scalability of our ORRAM device to nanoscale, we first fabricated the Pd bottom electrode with a width of 500 nm by electron-beam lithography. The MoO_x layer is deposited on top of Pd electrode. Then we use a hard mask to prepare ITO top

electrode, which forms a cross point with Pd electrode. Supplementary Fig. 2a presents the optical microscopy image and SEM image of Pd electrode. Supplementary Fig. 2b shows the I-V switching characteristics. The device can be SET from HRS to LRS by UV light illumination, and electrically RESET at a voltage of -2.2 V. The ON/OFF ratio is extracted as 31 at a read voltage of 0.1 V, showing comparable switching characteristics with the large MoO_x ORRAM device.

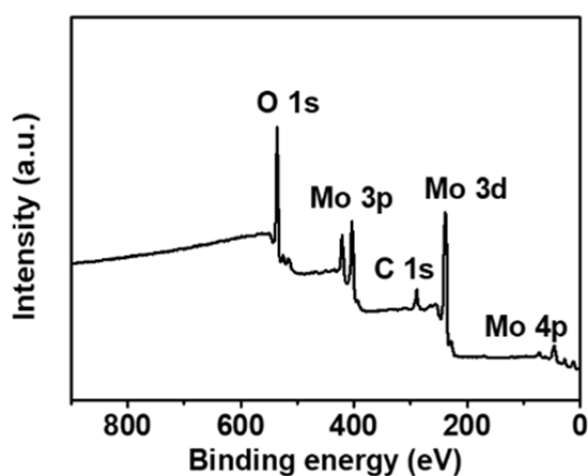
The cross-point device results suggest the scalability of bottom electrode to nanoscale. In order to further investigate the scalability of top electrode, we examined the I-V characteristics through conductive atomic force microscopy (C-AFM) measurements, where the Pt coated C-AFM tip works as the top electrode with a tip size of 14 nm (radius). The contact area is 150 nm². Supplementary Fig. 2c shows the schematic of the C-AFM measurement. MoO_x thin film is first prepared on Pd bottom electrode. The voltage is applied on the Pd electrode. The C-AFM measurement is conducted in a contact mode. Supplementary Fig. 2d shows the IV switching characteristics. Before the UV light illumination, the device is swept from 0 V to 4V, showing HRS. Then an *in-situ* UV light is illuminated to the samples. After the removal of the UV light illumination, another voltage sweep from 0 V to 4 V confirms the transition from HRS to LRS after the UV light illumination. As a negative voltage is applied on the Pd electrode, the HRS can be switched back to HRS again at -2.1 V. The C-AFM measurements confirm that the top electrode of our ORRAM device can be potentially scaled down to nanometer size.



Supplementary Fig. 2. Switching characteristics of ORRAMs with bottom electrode and top electrode scaled down to nanoscale. (a) Schematic drawing, optical image and SEM image of a cross-point device. (b) I-V characteristics of a cross-point ORRAM device. (c) Schematic of C-AFM measurement setup. (d) I-V characteristics of C-AFM measurement.

Supplementary Note 3

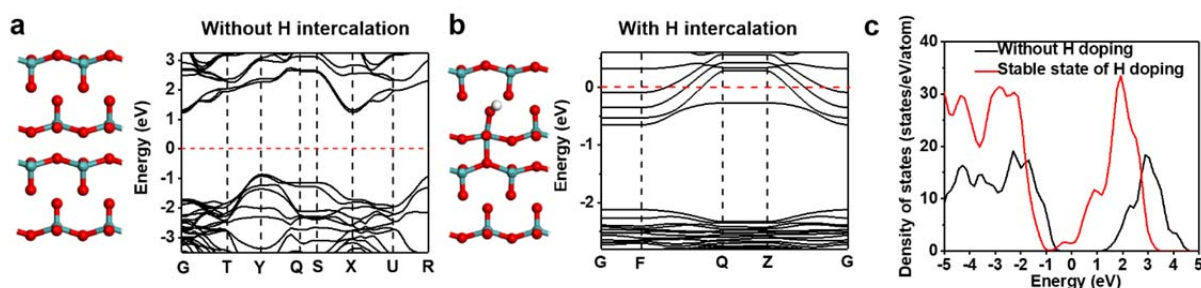
The full XPS spectrum of as-prepared MoO_x thin film is shown in Supplementary Fig. S3.



Supplementary Fig. S3. XPS full spectrum of as-prepared MoO_x layer on quartz substrate.

The band structure and density of states of α -MoO₃ bulk material before and after the proton intercalation (atomic ratio of H/Mo=1/8) are shown in Supplementary Fig. 4a and 4b, respectively. The density function theory (DFT) calculation suggests the transition from semiconducting MoO₃ to metallic bronze phase.²⁻⁴ The Fermi level shifts towards the conduction band after the intercalation of protons. The change of electronic structure can be attributed to the formation of H-bonding network and the charge transfer from H to O atoms. The highly polarized bonding environment causes the increased electron mobility and the metallic features in the hydrogen bronze.³

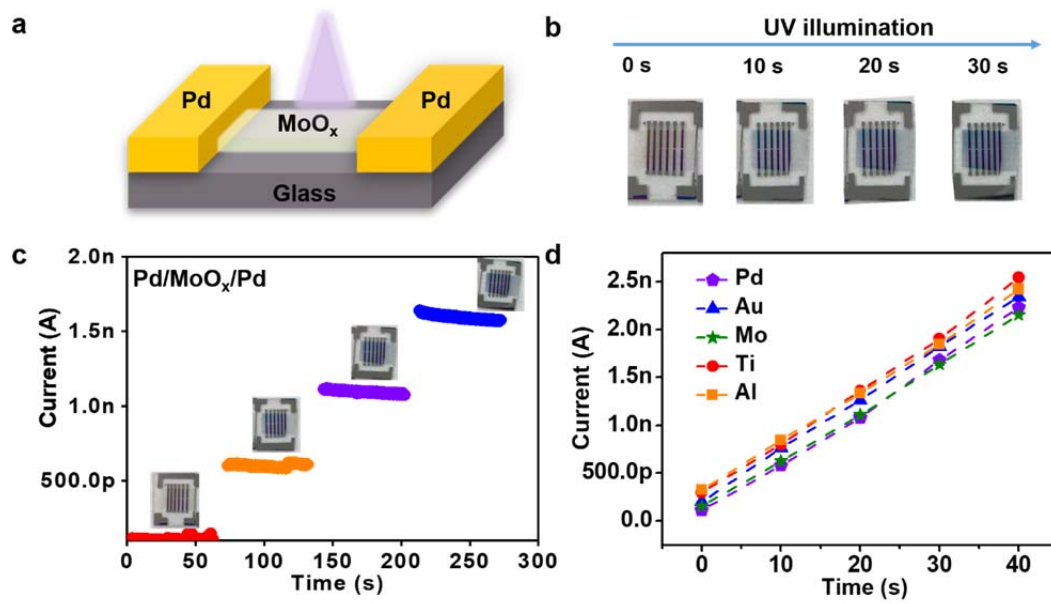
DFT computational methods: DFT calculations were performed through the Vienna ab initio simulation package (VASP) using the projector augmented wave (PAW) method with the generalized gradient approximation of Perdew–Burke–Ernzerhof (GGA-PBE) exchange correlation functional. Energy cutoff for plane-wave expansion was set to 480 eV. K-points grids of $15 \times 15 \times 15$ were chosen for the electronic structure calculations of bulk MoO₃ and H intercalated MoO₃. All the structures were fully relaxed using the conjugated gradient method until the Hellmann–Feynman force on each atom was less than 0.01 eV/Å.



Supplementary Fig. 4. DFT calculations. Atomic structures and band structures of α -MoO₃ bulk material at states of (a) without H intercalation. (b) stable state with H intercalation. (c) Density of states of MoO₃ bulk material with and without intercalation.

Supplementary Note 4

As the valence state change also corresponds to the colour change in the MoO_x film, we fabricated symmetric lateral devices on the glass substrate to observe the colour change of thin film under UV illumination. This colour change can directly associate the electrical conductivity of MoO_x with the color change/Mo valence state change. Supplementary Fig. 5a shows the schematics of the lateral devices with Pd as two electrodes. We firstly deposit patterned Pd electrodes with a channel length of 30 μm and a channel width of 70 μm on the glass substrate. MoO_x film was then deposited on top of the patterned electrodes and glass substrate. Supplementary Fig. 5b reveals the gradual colour change of the device with UV illumination time. Supplementary Fig. 5c records the corresponding device resistances at different illumination times (0s, 10 s, 20 s, 30 s). It can be clearly observed that the electrical conductivity of the MoO_x device increases gradually with the colour changing from transparent to light blue under the UV light illumination. Other metal electrodes with different work functions, such as Au, Mo, Ti, Al, are also employed. Supplementary Fig. 5d demonstrates the electrical conductivity as a function of UV illumination time. The increasing rates of conductivity are similar for the devices with different metal electrodes, which further suggests the conductivity change originating from the MoO_x layer itself instead of the interface between MoO_x and metal electrodes.



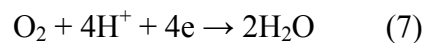
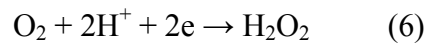
Supplementary Fig. 5. MoO_x lateral device. (a) Schematic of lateral MoO_x device with Pd as electrodes. (b) Colour change of the MoO_x lateral devices during continuous UV light illumination. During the illumination process, the colour gradually changes from transparent to blue. (c) Four resistance states corresponding to illumination times of 0 s, 10 s, 20 s and 30 s, respectively. (d) Variations of device conductance with illumination time for devices with different metal electrodes.

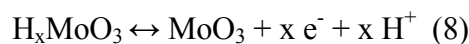
Supplementary Note 5

Accompanying with the valence state change in Mo ions, the resistance of the device will also change from HRS to LRS. The process can be expressed as:

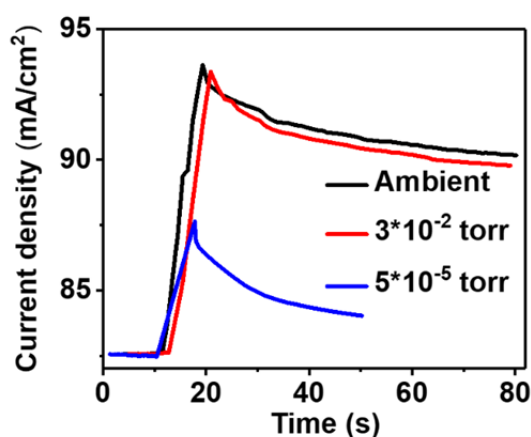


During the electrical RESET process, protons is extracted from MoO₃, and drifts towards the Pd bottom electrode. Pd is a well-accepted metal catalyst for oxygen reduction reaction (ORR). As protons are extracted from MoO_x layer, the reverse reaction from H_yMoO_x to MoO_x is promoted, corresponding to HRS:





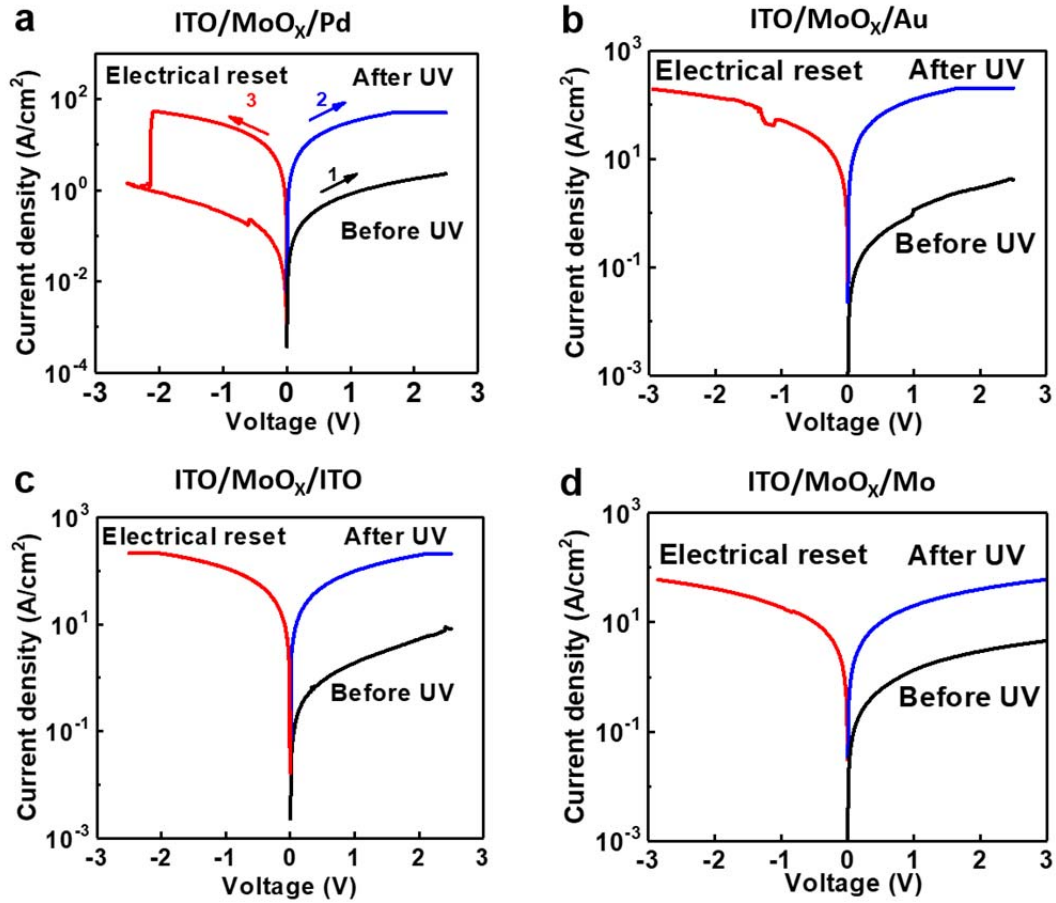
The produced H_2O_2 or H_2O in the ORR can be cyclically used for the UV SET process. Under the UV illumination, H_2O_2 is dissociated into H_2O and O_2 . The produced H_2O can be also used for the water dissociation process under the UV illumination. Therefore, during the cyclic process, only small amount of water is required. To investigate the influences of water on the performances, the device was characterized under three different ambient vapour pressures: 750 Torr, 3×10^{-2} Torr and 5×10^{-5} Torr, respectively (Supplementary Fig. 6). The switching characteristics are similar under 750 Torr and 3×10^{-2} Torr; as the pressure is further reduced to 5×10^{-5} Torr, an obvious current drop can be observed. These results suggest that the amount of water required for the cyclic SET processes is very few.⁵



Supplementary Fig. 6. Current responses as a function of time under different ambient water-vapor pressure. The light illumination lasts for 10 s with the power density of 7.6 mW/cm^2 .

To further confirm the role of Pd electrode, we also compare the switching characteristics of devices with different bottom electrodes with no/weak catalytic activity in Supplementary Fig. 7: ITO/MoO_x/Pd, ITO/MoO_x/ITO, ITO/MoO_x/Au and ITO/MoO_x/Mo. All the four devices can be SET by UV illumination, resulting from the change in valence states of Mo in the MoO_x thin film. However, the devices with bottom electrodes of Mo, ITO and Au cannot be RESET to HRS, in sharp contrast to the Pd/MoO_x/ITO cell. These results further suggest

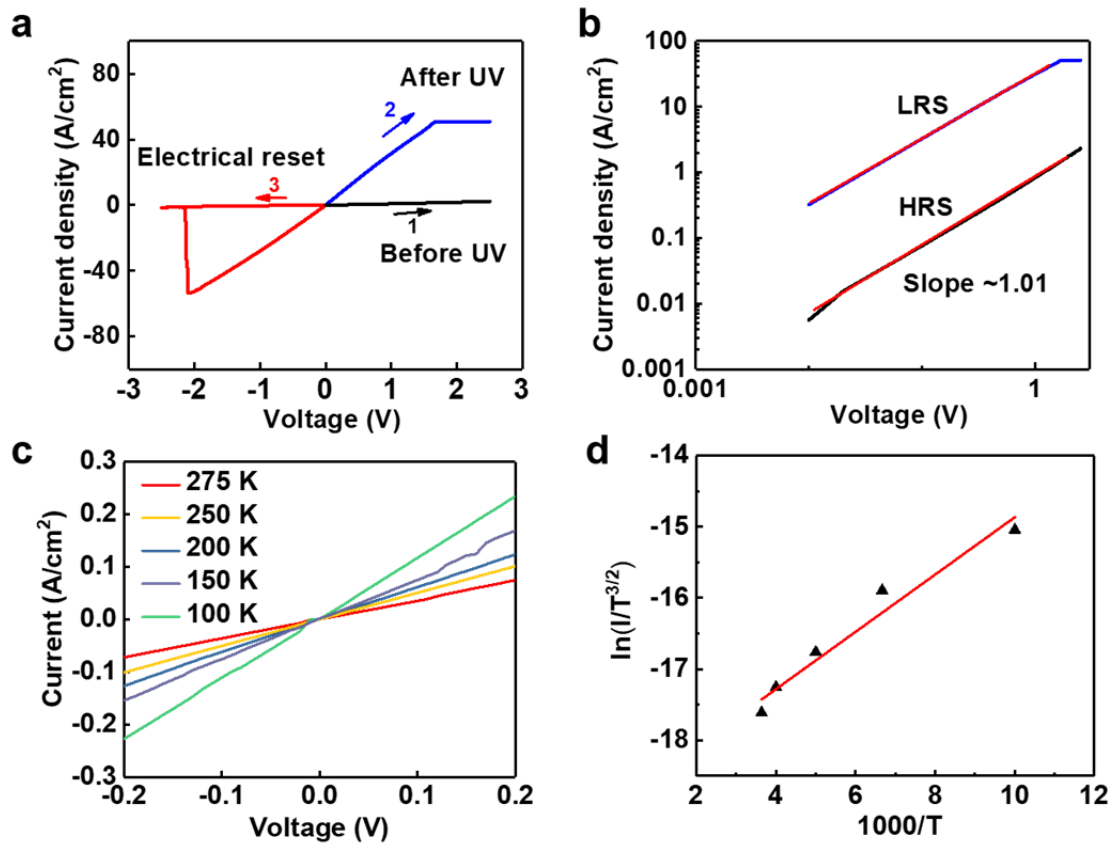
the catalytic activity of Pd in ORR, which consumes H^+ and promote the reverse reaction from H_yMoO_x to MoO_x .



Supplementary Fig. 7. Comparisons of switching characteristics of MoO_x devices with different bottom electrodes: Au, ITO and Mo.

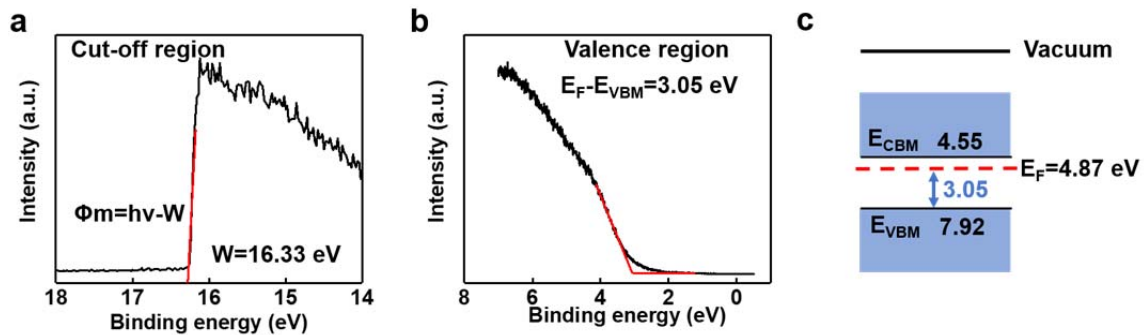
In addition, we also performed control experiments to verify that the modulation of ITO/ MoO_x interface Schottky barrier is not the dominate switching mechanism in the ORRAM device. First, we plot the IV switching curve in linear scale of the Pd/ MoO_x /ITO ORRAM device (Supplementary Fig. 8a). For both HRS and LRS, the IV curves exhibit linear relationship. In addition, double log I-log V plots of HRS and LRS are shown in Supplementary Fig. 8b. The slopes of the fitting curves are extracted as ~ 1.01 for both HRS and LRS, which suggest the Ohmic conduction at both HRS and LRS. To further verify the

Ohmic contact between ITO and MoO_x , we also conducted temperature-dependent IV measurements of ITO/ MoO_x /ITO structure, as shown in Supplementary Fig. 8c. The IV curves still exhibit linear relationships with the temperature from 100 K to 275 K, which suggests the Ohmic contact in a wide range of temperatures. Arrhenius plot of $\ln(I/T^{3/2})$ versus $1000/T$ is illustrated in Supplementary Fig. 8d. The positive slope indicates nearly zero Schottky barrier in the ITO/ MoO_x contacts. From the temperature-dependent measurements, the modulation of ITO/ MoO_x interface barrier can be considered as not the dominate switching mechanism in the ORRAM device.



Supplementary Fig. 8. Temperature-dependent electrical characteristics. (a) Linear plot of IV switching curve in the Pd/ MoO_x /ITO ORRAM device at room temperature. (b) Double log plot of IV switching curve in the Pd/ MoO_x /ITO ORRAM device at both HRS and LRS at room temperature. (c) Low-temperature IV measurements of ITO/ MoO_x /ITO structure with temperature from 100 K to 275 K. (d) Arrhenius plot of $\ln(I/T^{3/2})$ versus $1000/T$.

Furthermore, we also determine the work function and band diagram of as-prepared MoO_x thin film from the ultraviolet photoelectron spectroscopy (UPS) test, as shown in Supplementary Fig.9. A He-I UV light source ($h\nu = 21.2$ eV) was utilized to obtain the UPS spectra. The binding energy is referred to the Fermi level. We extract the work function from the onset energy in the cut-off region, as shown in Supplementary Fig. 9a, which is 4.87 eV. Supplementary Fig. 9b shows the spectrum near the valence band region, where the valence band maxima is obtained at a binding energy of 1.32 eV. The band gap of MoO_x thin film is determined to be 1.63 eV, according to the absorption spectrum in Supplementary Fig. 1b. Thus, the schematic band diagram of MoO_x thin film can be sketched in the inset of Supplementary Fig. 9c, in which the Fermi level locates near to the conduction band. The work function of MoO_x thin film (4.87 eV) is close to that of ITO (~ 4.7 eV)^{6,7}, which also suggests a small Schottky barrier between MoO_x and ITO. Thus, we propose that the dominate switching mechanism is considered as the valence state transition of Mo ions in our ORRAM device.



Supplementary Fig. 9. UPS test of as-prepared MoO_x on Pd substrate. (a) Cut-off region. The work function is extracted as 4.87 eV. (b) Valence region. The difference between Fermi level and valence band maximum is extracted as 3.05 eV. (c) Band diagram of as-prepared MoO_x film.

Supplementary Note 6

We examined the switching characteristics of ORRAM devices with different MoO_x thicknesses (50 nm, 100 nm, 400 nm and 800 nm) in Supplementary Fig. 10. All the

ORRAM devices can be optically SET through UV light illumination (365 nm/150 mW/cm²). The ON/OFF ratios for the ORRAMs with four MoO_x thicknesses (50 nm, 100 nm, 400 nm and 800 nm) are 32, 41, 33 and 38 at a read voltage of 0.1 V, respectively, as documented in Supplementary Table I. It can be observed that the ON/OFF ratios are similar for the MoO_x devices with different thicknesses. Although thicker MoO_x layer can provide larger amount of water molecules and Mo⁶⁺, the proportion of Mo⁶⁺/Mo⁵⁺ transition per unit volume is relatively independent on the thickness. This thickness-independent characteristic of ON/OFF ratio can be explained from the following equations:

The resistances at LRS and HRS can be expressed as:

$$R_{LRS} = \frac{V}{I_{LRS}}; I_{LRS} \propto n(Mo^{5+})_{LRS} \times d \quad (9)$$

$$R_{HRS} = \frac{V}{I_{HRS}}; I_{HRS} \propto n(Mo^{5+})_{HRS} \times d \quad (10)$$

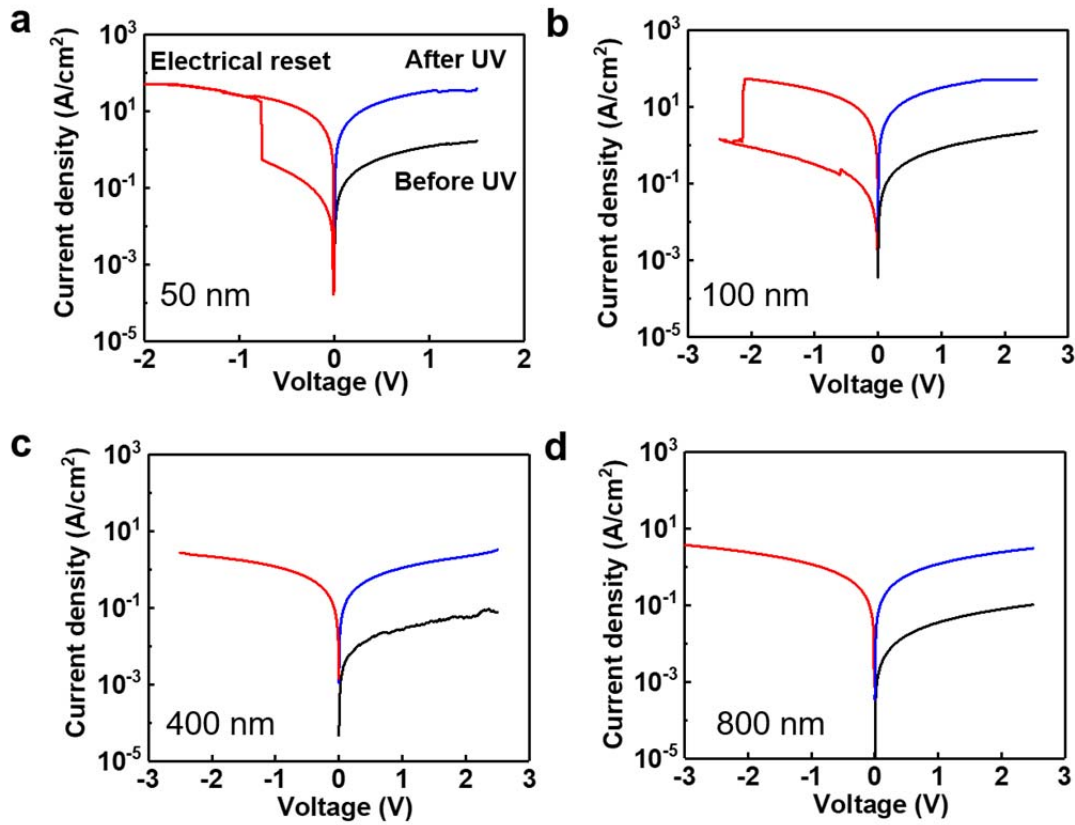
Where $n(Mo^{5+})$ is the number of Mo⁵⁺ per unit volume, d is the thickness of MoO_x layer, thus the ON/OFF ratio satisfies the relationship as below:

$$\frac{R_{HRS}}{R_{LRS}} = \frac{I_{LRS}}{I_{HRS}} \propto \frac{n(Mo^{5+})_{LRS}}{n(Mo^{5+})_{HRS}} \quad (11)$$

According to the equations (9)-(11), the ORRAM ON/OFF ratio is proportional to the amount of Mo⁶⁺ to Mo⁵⁺ transition per unit volume. As the amount of Mo⁶⁺ to Mo⁵⁺ transition per unit volume for the MoO_x with different thicknesses is relatively close under similar light illumination, the ON/OFF ratios in the optical SET process exhibit nearly no thickness-dependent characteristics.

For the electrical RESET process, it is very difficult to electrically RESET the ORRAMs with MoO_x thicknesses of 400 nm and 800 nm. During the electrical RESET process, protons

are extracted from MoO_x , and drift towards the Pd bottom electrode, which serves as proton reservoir. As protons are extracted from MoO_x layer, the reverse reaction from H_yMoO_x to MoO_x is promoted ($\text{H}_y\text{MoO}_x \leftrightarrow \text{MoO}_x + y\text{e}^- + y\text{H}^+$). For the ORRAM devices with thicker MoO_x layer, the diffusion lengths of protons are much longer, thus requiring larger driving force for drifting protons towards Pd electrode. As a result, the RESET process for the ORRAMs with 400 nm and 800 nm thick MoO_x layer can hardly be observed. For the ORRAMs with thinner MoO_x films, the RESET voltage of ORRAM with 50 nm MoO_x (0.8 V) is smaller than that with 100 nm MoO_x (2.1 V). It can also be associated with the shorter drifting length of protons in thinner MoO_x , and thus the smaller driving force.



Supplementary Fig. 10. Resistive switching characteristics of ORRAM devices with different MoO_x thicknesses. (a) 50 nm, (b) 100 nm, (c) 400 nm and (d) 800 nm. The black line and blue line show the IV sweep before and after UV illumination, respectively. The red line shows the electrical reset process.

Supplementary Table I. ON/OFF ratios of ORRAMs with different MoO_x thicknesses.

MoO _x thickness (nm)	50	100	400	800
ON/OFF ratio	32	41	33	38

Supplementary Note 7

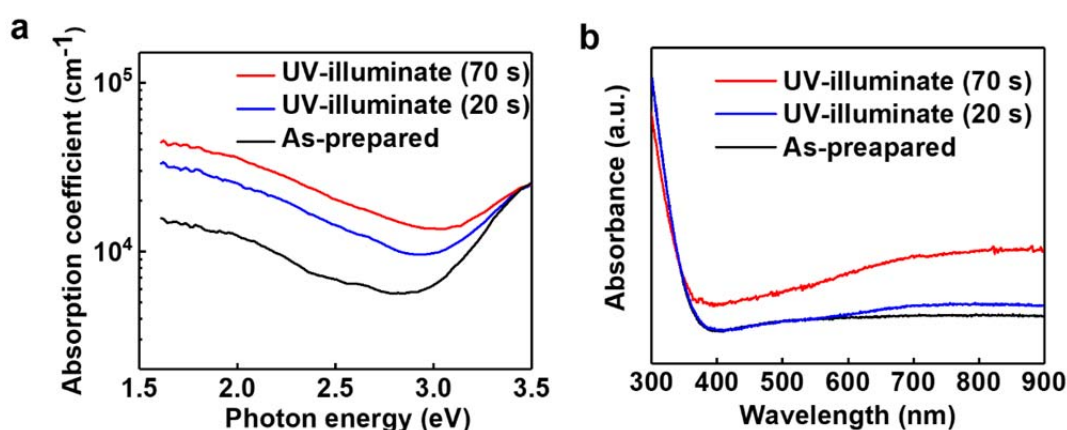
Photothermal deflection spectroscopy (PDS) is employed to examine the density of band gap states of three samples (different UV light illumination time) prepared on quartz substrates, as shown in Supplementary Fig. 11a. PDS test can be used to evaluate defect densities of thin films with high sensitivity (order of 10^{-4} cm^{-1}) through absorption characterization. We characterized the three samples with different light illumination time durations, *i.e.*, as-prepared MoO_x thin film, UV (365 nm)-illuminated MoO_x thin film for 20 s, and UV (365 nm)-illuminated MoO_x thin film for 70 s, respectively. The light intensity used for the illumination is 0.88 mW/cm^2 . The PDS spectra shows the absorption coefficient increases with UV illumination duration within the photon energy from 1.6 eV to 3.5 eV. The absorption coefficient $\alpha(E)$ is proportional to the rate of optical transitions from all initially occupied states N_i to all possible unoccupied final states N_f separated by an energy of $E=h\nu$. The relationship between absorption coefficient and the defect sates can be expressed as ⁸:

$$\alpha(E) = \left(\frac{\text{const}}{E} \right) \int N_i(\varepsilon) f(\varepsilon) N_f(\varepsilon + E) [1 - f(\varepsilon + E)] d\varepsilon \quad (13)$$

$f(\varepsilon)$ is the appropriate occupation function. Thus, the increased absorption coefficient suggests the sub-gap absorption, corresponding to the defect density, increases with the UV illuminated time.

In addition to PDS characterization, UV-Vis (Ultraviolet visible spectroscopy) spectra of MoO_x thin films also roughly exhibit the similar trend (Supplementary Fig. 11b). As the UV light is illuminated, the absorption in the visible region is stronger compared to the as-

prepared sample, which also confirms more defect states induced with longer light illumination time. The presence of more defect states in the MoO_x thin film can give rise to more stable formation of Mo^{5+} . According to the proposed reaction equations (3)-(5), with more defect states in the film, more photo-generated electrons can be trapped in the defect states, which promotes the water dissociation reaction and more stable formation of protons and Mo^{5+} .



Supplementary Fig. 11. Characterisations of defect states before and after UV illumination. (a) PDS spectra and (b) absorption spectra of as-prepared MoO_x thin film, as-prepared MoO_x thin film, UV-illuminated MoO_x thin film for 20 s and UV-illuminated MoO_x thin film for 70 s, respectively.

Supplementary Note 8

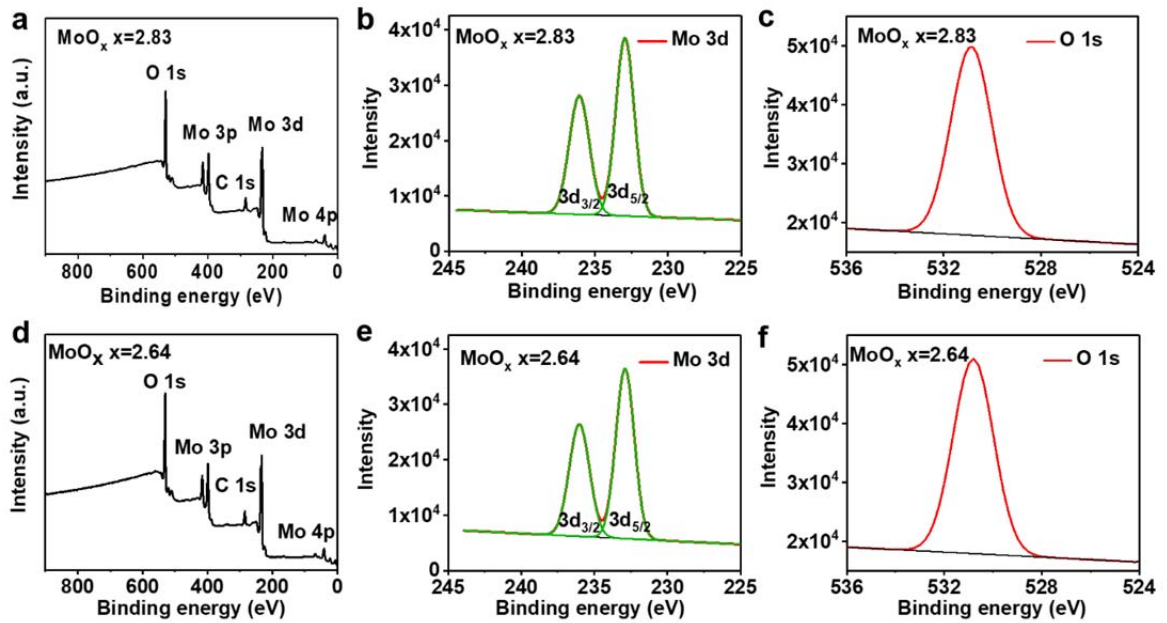
In addition to 365 nm UV light, our device can also respond to blue light (410 nm) as we can tune the bandgap of the MoO_x thin films. We also investigate the effect of stoichiometry of MoO_x on its bandgap. Engineering the bandgap by materials stoichiometry potentially enables the response to visible light.⁹

We first performed spectroscopy studies on the MoO_x thin films (x values of 2.83 and 2.64, respectively) to reveal the relationship between stoichiometry and bandgap. Supplementary Fig. 12 shows the XPS spectra and narrow scans of Mo 3d peaks and O1s peak of the MoO_x thin films with two different compositions. According to the narrow scans of Mo 3d peaks

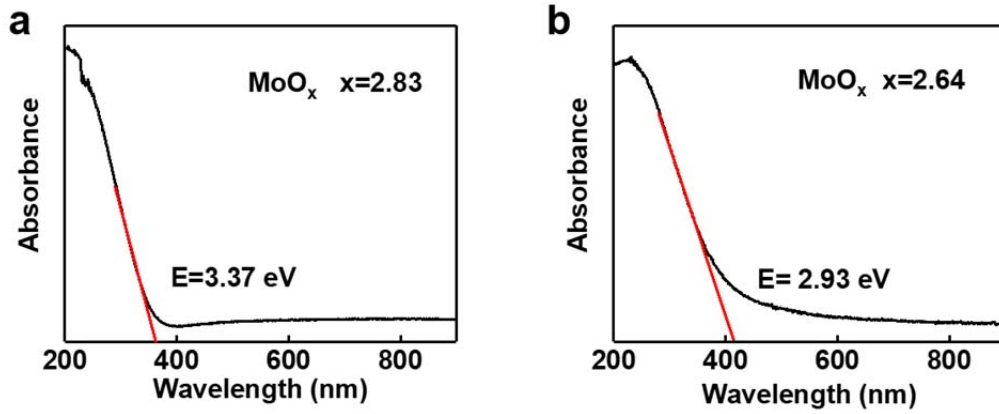
and O1s peak, the atomic ratio of Mo/O in MoO_x can be calculated according to Equation (14):

$$\frac{n_{\text{Mo}}}{n_{\text{O}}} = \frac{\frac{I_{\text{Mo } 3d_{5/2}}}{S_{\text{Mo } 3d_{5/2}}}}{\frac{I_{\text{O } 1s}}{S_{\text{O } 1s}}} \quad (14)$$

where $I_{\text{Mo } 3d_{5/2}}$ is the peak height of Mo $3d_{5/2}$ peak, $S_{\text{Mo } 3d_{5/2}}$ is the atomic sensitivity factor of Mo $3d_{5/2}$, $I_{\text{O } 1s}$ is the peak height of O1s peak, and $S_{\text{O } 1s}$ is the atomic sensitivity factor of O1s. The corresponding absorption spectra of different x values are shown in Supplementary Fig. 13. As the x value is reduced from 2.83 to 2.64, the band gap of MoO_x is reduced from 3.37 eV to 2.93 eV.

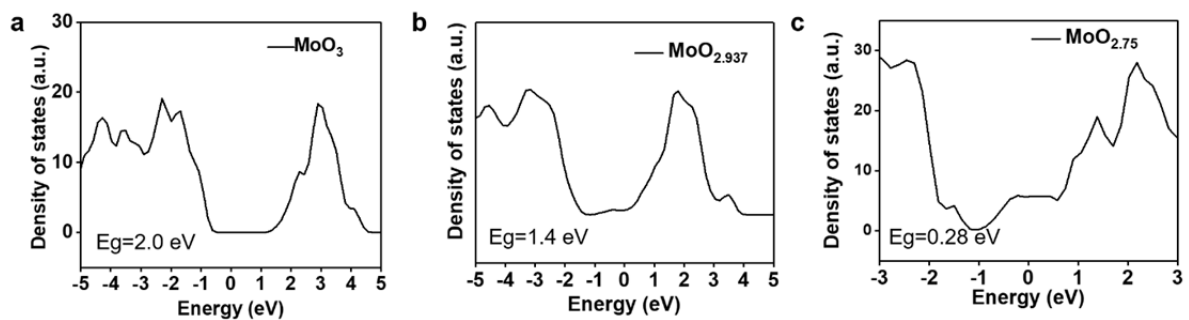


Supplementary Fig. 12. XPS spectra of MoO_x thin film. $x=2.83$: (a) Wide scan and (b) Narrow scan of Mo 3d peaks. (c) Narrow scan of O1s peak. $x=2.64$: (d) Wide scan and (e) Narrow scan of Mo 3d peaks. (f) Narrow scan of O1s peak.



Supplementary Fig. 13. Absorption spectra of MoO_x thin film with x value of 2.83 and 2.64, respectively.

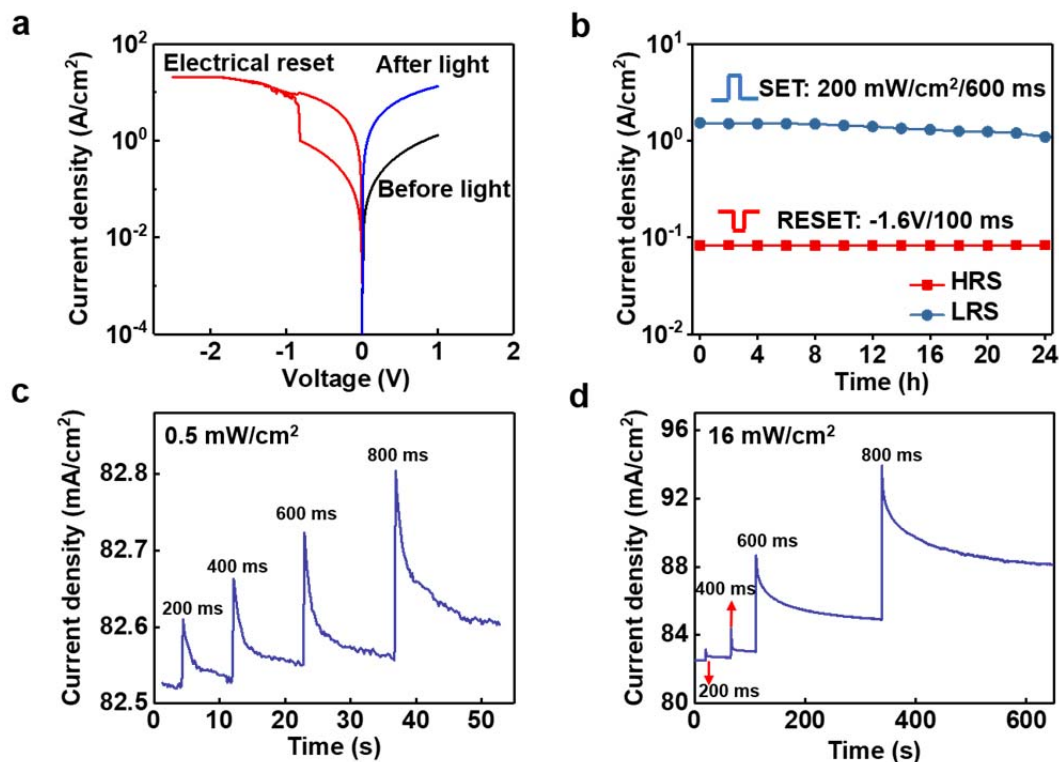
We also conducted density function theory (DFT) calculations on the bandgap of the MoO_x with different stoichiometry, which suggests the band gap of MoO_x is decreased as value of x is reduced, as shown in Supplementary Fig. 14. The band gap for MoO₃ is extracted as 2.0 eV. As x value is decreased to 2.937 and 2.75, the band gap is reduced to 1.4 eV and 0.28 eV, respectively. From the DFT calculation, it suggests that it is possible to reduce the band gap of MoO_x by decreasing x value in the MoO_x thin film, which is in agreement with the previous report.⁹ This characteristic also open up a possibility to respond to visible light.



Supplementary Fig. 14. DFT simulations. Density of states (DOS) in MoO_x thin films with different x values of (a) 3, (b) 2.937 and (c) 2.75 calculated by density function theory, respectively.

As the band gap of MoO_x thin film can be tuned to 2.93 eV, which enables the absorption of blue light, we then illuminated blue light with the wavelength of 410 nm to operate the

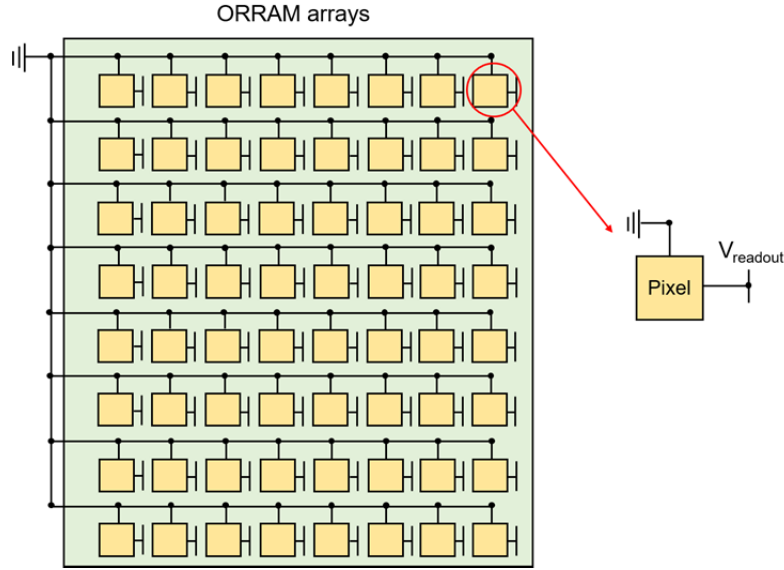
ORRAM device. Supplementary Fig. 15a shows the IV curve of non-volatile switching excited by blue light. The top ITO electrode was grounded, and the Pd bottom electrode was connected to an external voltage source. Supplementary Fig. 15a shows the IV curve of ORRAM device with a voltage sweep from 0 V to 1 V before the blue light illumination (black line), suggesting the ORRAM device is at a high resistance state (HRS). Then blue light with wavelength of 410 nm and power density of 200 mW/cm² was illuminated to the device. After the removal of blue light, a voltage is swept from 0 V to 1 V (blue line), exhibiting that the device is switched to a low resistance state (LRS). The device then can be electrically reset by applying a negative voltage sweep. A RESET process occurs at -0.8 V. Both HRS and LRS can be retained for 24 h after the removal of both optical and electrical stimulation, as shown in Supplementary Fig. 15b, where an optical pulse (410 nm) with a power density of 200 mW/cm² and a pulse width of 600 ms is employed for the SET process, and an electrical pulse with voltage of -1.6 V and pulse width of 100 ms is employed for the RESET process. In addition to the non-volatile switching characteristics, we can also demonstrate volatile switching characteristics by applying the blue light with lower intensity, as shown in Supplementary Fig. 15c and Supplementary Fig. 15d. The ORRAM device can also respond to different light dosage including light intensity and illumination time, which is similar to the UV light excitation. Supplementary Fig. 15c and Supplementary Fig. 15d illustrate the illumination time-dependent current responses of ORRAM device under blue light illuminations with two different intensities of 0.5 mW/cm² and 16 mW/cm², respectively. It is observed that longer illumination time or higher light intensity results in higher current and longer retention time, in consistent with the previous measurement under the illumination of UV light. Our ORRAM device can respond to the UV light and the blue light in the visible light region.



Supplementary Fig. 15. Switching characteristics of ORRAM device with 410 nm blue light excitation. (a) Non-volatile switching characteristics. The black line is the voltage sweep before light illumination from 0 V to 1 V. The blue line shows the voltage sweep after light illumination from 0 V to 1 V. (b) Retention properties with pulse switching. The optical pulse with light intensity of 200 mW/cm^2 and pulse width of 600 ms is employed for SET process. The electrical pulse with voltage of -1.6 V and pulse width of 100 ms is employed for RESET process. (c) Current responses excited optical pulses with different illumination time at light intensity of 0.5 mW/cm^2 . (d) Current responses excited optical pulses with different illumination time at light intensity of 16 mW/cm^2 .

Supplementary Note 9

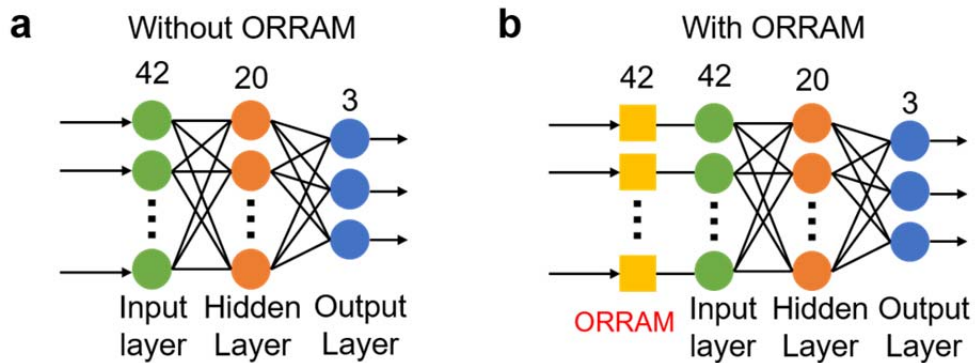
The readout circuit of the 8×8 ORRAM array is illustrated in Supplementary Fig. 16. The current readout process was performed one cell by one cell, similar to conventional image sensor.



Supplementary Fig. 17. Readout circuit of ORRAM arrays. The generated current is read out with a voltage of 0.1 V.

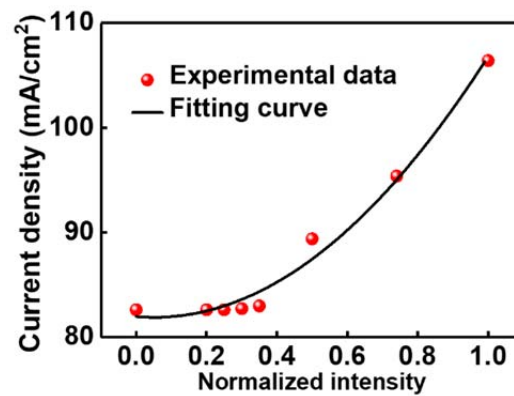
Supplementary Note 10

Supplementary Supplementary Fig. 17a shows the $42 \times 20 \times 3$ three-layered artificial neural network we employed in this work. Back-propagation (BP) algorithm is used for training and recognition processes. The activation function in Hidden layer and Output layer is Sigmoid. The learning rate is set as a fix value 0.01. For the structure of neural network connected with ORRAM (Supplementary Supplementary Fig. 17b), ORRAMs are connected to the input layer directly in the neural network. In the training process, images are exposed to the ORRAM or delivered to the input layer one by one.



Supplementary Fig. 17. Illustration of artificial neural network. (a) The structure of the three-layered artificial neural network without ORRAM. (b) The structure of the three-layered artificial neural network with ORRAM.

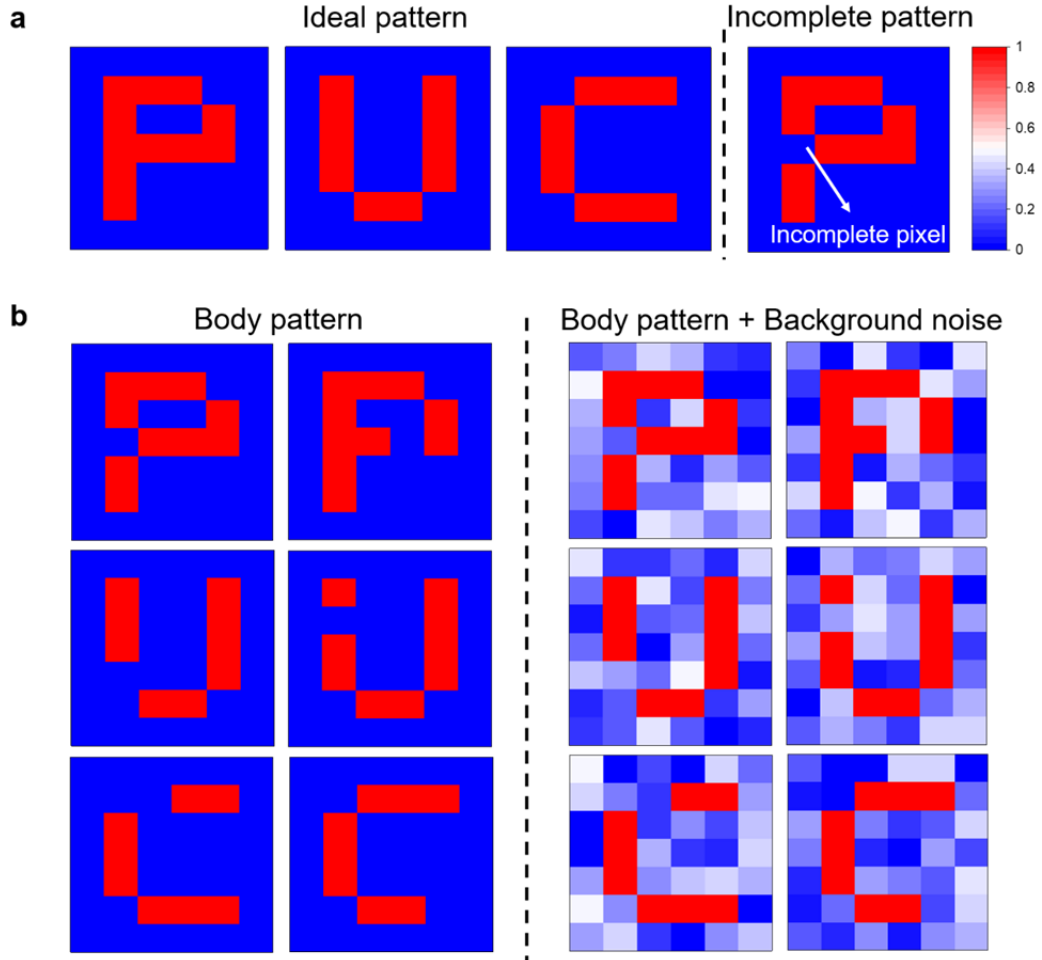
The fitting curve of current density as a function of normalized light intensity is shown in Supplementary Supplementary 18. The stimuli used are 100 pulses with pulse width of 200 ms and pulse interval of 600 ms. The background noise signals in the simulation are randomly generated in a range of 0-0.5 (normalized intensity). The corresponding output currents through ORRAM devices are determined according to the fitting curve.



Supplementary Fig. 18. Fitting curve of current density vs. normalized light intensity. 100 pulse stimulations with pulse width of 200 ms and pulse interval of 600 ms are used. The current densities are extracted after 100 s of the stimulation.

The image dataset in the simulation are generated through two steps. The body patterns of the image dataset are composed of three letters (P, U, C) with different incomplete pixels. The ideal pattern of three letters and examples of an image of letter P with an incomplete pixel are shown in Supplementary Fig. 19a. Supplementary Fig. 19b presents the examples of the images with body patterns and the images with both body patterns and background noise signals. The background noise signals correspond to different optical signals from the real image. Excluding the body pattern of the letters, the noise signals are randomly generated by a computer program by MATLAB for the other pixels in the image. As the images are

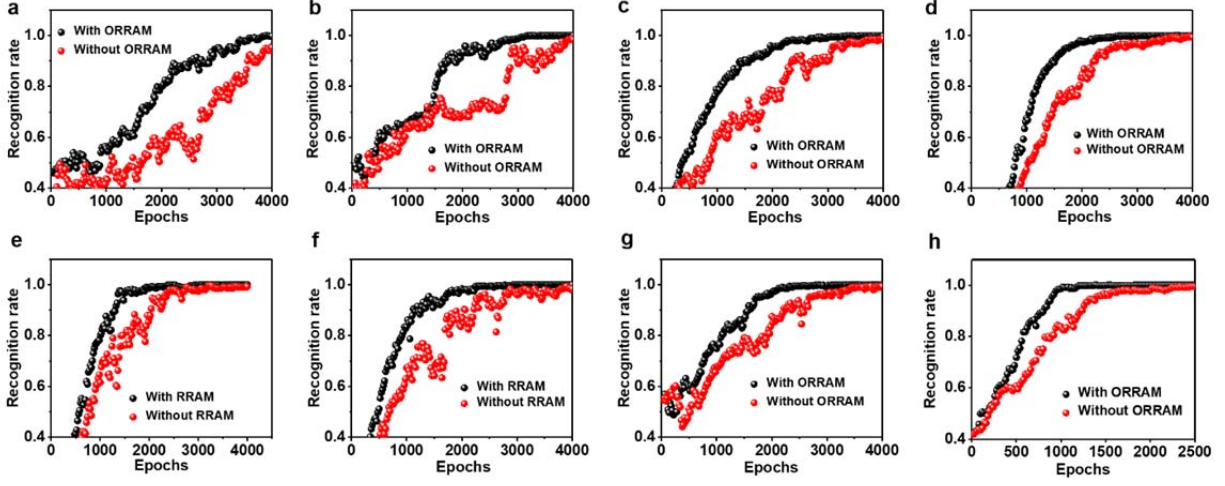
presented to the ORRAM arrays (one image pixel corresponds to one ORRAM device), ORRAM arrays convert optical signals to output currents. The output currents are then reshaped into a new output image.



Supplementary Fig. 19. Demonstrations of examples of image dataset and background noise reduction in the images through ORRAM devices. (a) Idea patterns of three letters (P, U, C) and examples of pattern P with incomplete pixel. The image dataset is defined with different patterns of three letters (P, U, C) with incomplete pixel. (b) On the basis of body pattern, the background noise signals are assigned randomly to each image.

Supplementary Fig. 20 show the comparisons of recognition rate in visual system with and without ORRAM at different 8 simulation running cycles. The background noise signals are generated randomly in 8 running cycles. For the image database with different background noise, the recognition rates with ORRAM are always higher than that without ORRAM

before the stabilization; and the training efficiencies are higher with ORRAM based pre-processing, which suggests the repeatability of the simulation.



Supplementary Fig. 20. Comparisons of recognition rate at 6 different simulation running cycles. The variation of recognition rate with increasing training epochs with (black line) and without (red line) ORRAM for pre-processing, respectively.

Considering from the improvements of efficiency (in energy and speed) in classification tasks with the employment of ORRAMs, we extract the epochs required for different recognition accuracies with and without ORRAM from the simulation. Due to the structures of neural network with and without ORRAM are similar, single epoch has approximately equal energy consumption and speed. Therefore, for the same accuracy, the whole processing speed is inversely proportional to the epochs, and the whole energy consumption is proportional to the epochs. The results are shown in Supplementary Supplementary II. Rate of improvement is defined as:

Rate of improvement (%)

$$= \frac{\text{Epochs (without ORRAM)} - \text{Epochs (with ORRAM)}}{\text{Epochs (without ORRAM)}} \times 100\% \quad (15)$$

The value of epochs is the average value for 10 independent learning and recognition processes. To achieve a recognition accuracy of 99%, it requires 3740 epochs for recognizing

the images without ORRAM; after the employment of ORRAM array for pre-processing, it requires 2186 epochs (41.5% reduction compared with the epochs required without ORRAM). With the same image database in our simulation work, the image recognition task with ORRAM is more efficient in terms of both speed and energy consumption compared with the classification without ORRAM. Supplementary Table II shows the relationship between recognition accuracy and epochs with and without ORRAM pre-processing. The rate of improvements exhibits substantial increase.

Supplementary Table II. Epochs for the classification tasks with and without ORRAM at different recognition accuracies. The rate of improvements in epochs with the employment of ORRAM are also extracted in the table.

Recognition accuracy	Without ORRAM (Epochs)	With ORRAM (Epochs)	Rate of improvements
0.80	1216	812	33.2%
0.90	2000	1488	25.6%
0.95	2205	1634	25.8%
0.98	2571	1771	31.1%
0.99	3740	2186	41.5%

References

- 1 Ajito, K., Nagahara, L., Tryk, D., Hashimoto, K. & Fujishima, A. Study of the photochromic properties of amorphous MoO₃ films using Raman microscopy. *J. Phys. Chem.* **99**, 16383-16388 (1995).
- 2 Lei, Y.-H. & Chen, Z.-X. DFT+ U study of properties of MoO₃ and hydrogen adsorption on MoO₃ (010). *J. Mater. Chem. C* **116**, 25757-25764 (2012).
- 3 Sha, X., Chen, L., Cooper, A. C., Pez, G. P. & Cheng, H. Hydrogen absorption and diffusion in bulk α -MoO₃. *J. Mater. Chem. C* **113**, 11399-11407 (2009).
- 4 Scanlon, D. O. et al. Theoretical and experimental study of the electronic structures of MoO₃ and MoO₂. *J. Mater. Chem. C* **114**, 4636-4645 (2010).
- 5 Tsuruoka, T. et al. Effects of Moisture on the Switching Characteristics of Oxide-Based, Gapless □ Type Atomic Switches. *Adv. Funct. Mater.* **22**, 70-77 (2012).

- 6 Parker, I. D. Carrier tunneling and device characteristics in polymer light-emitting diodes. *J. Appl. Phys.* **75**, 1656-1666 (1994).
- 7 Balasubramanian, N. & Subrahmanyam, A. Studies on evaporated indium tin oxide (ITO)/silicon junctions and an estimation of ITO work function. *J. Electrochem. Soc.* **138**, 322-324 (1991).
- 8 Fritzsche, H. *Amorphous Silicon and Related Materials*. (World Scientific, 1989).
- 9 Inzani, K. et al. Electronic properties of reduced molybdenum oxides. *Phys. Chem. Chem. Phys.* **19**, 9232-9245 (2017).